

F3 Definition of logarithm

Name: _____

Class: _____

Date: _____

Time: **343 minutes**

Marks: **287 marks**

Comments:

EXAM PAPERS PRACTICE

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Q1.

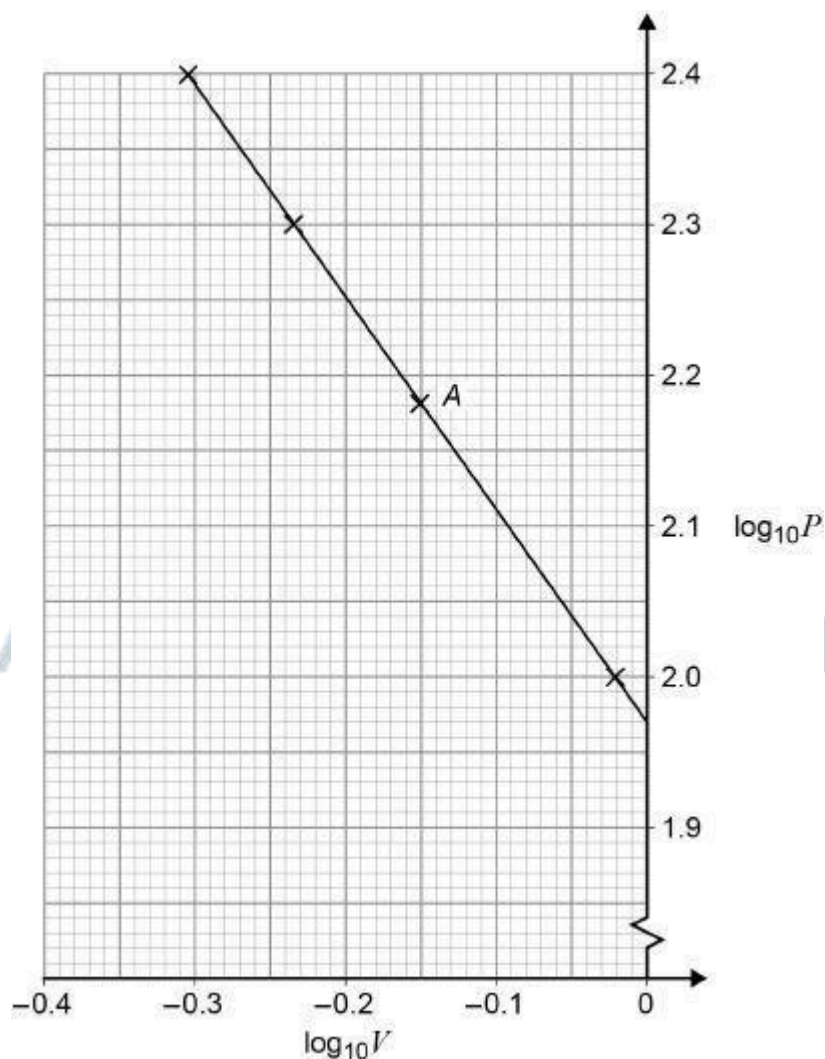
Maxine measures the pressure, P kilopascals, and the volume, V litres, in a fixed quantity of gas.

Maxine believes that the pressure and volume are connected by the equation

$$P = cV^d$$

where c and d are constants.

Using four experimental results, Maxine plots $\log_{10} P$ against $\log_{10} V$, as shown in the graph below.



- (a) Find the value of P and the value of V for the data point labelled A on the graph.

(2)

- (b) Calculate the value of each of the constants c and d .

(4)

- (c) Estimate the pressure of the gas when the volume is 2 litres.

(2)

(Total 8 marks)

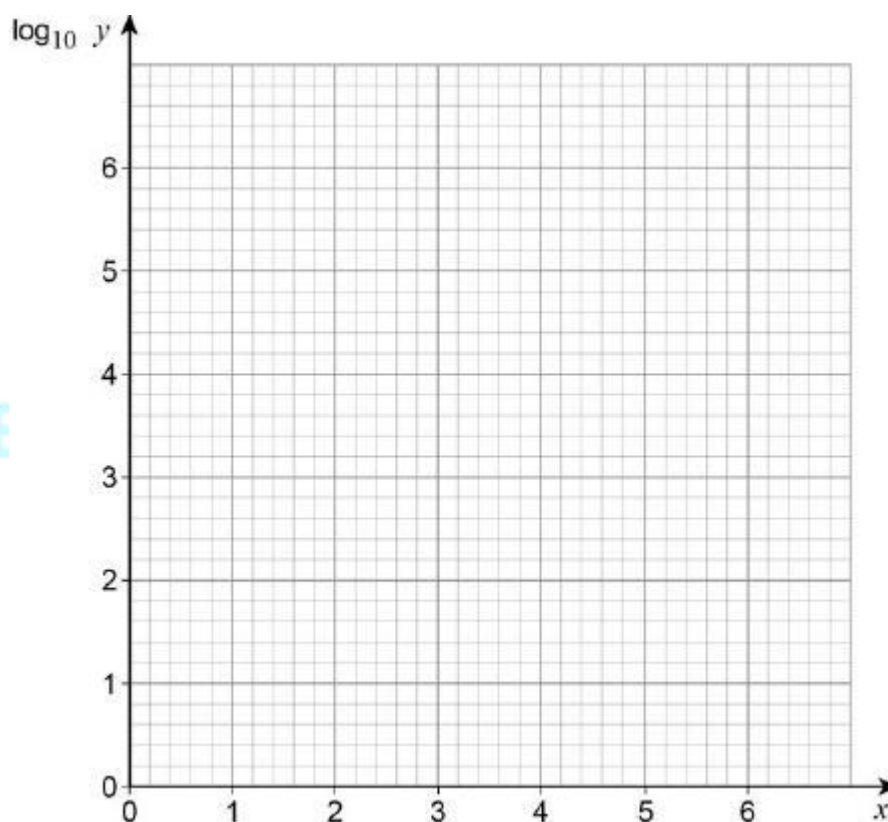
Q2.

A student conducts an experiment and records the following data for two variables, x and y .

x	1	2	3	4	5	6
y	14	45	130	1100	1300	3400
$\log_{10} y$						

The student is told that the relationship between x and y can be modelled by an equation of the form $y = kb^x$

- (a) Plot values of $\log_{10} y$ against x on the grid below.



(2)

- (b) State, with a reason, which value of y is likely to have been recorded incorrectly.

(1)

- (c) By drawing an appropriate straight line, find the values of k and b .

(4)

(Total 7 marks)

Q3.

Find the value of $\log_a(a^3) + \log_a\left(\frac{1}{a}\right)$

(Total 2 marks)

Q4.

A curve has equation $y = e^{2x}$

Find the coordinates of the point on the curve where the gradient of the curve is $\frac{1}{2}$
 Give your answer in an exact form.

(Total 5 marks)

Q5.

David has been investigating the population of rabbits on an island during a three-year period.

Based on data that he has collected, David decides to model the population of rabbits, R , by the formula

$$R = 50e^{0.5t}$$

where t is the time in years after 1 January 2016.

(a) Using David's model:

(i) state the population of rabbits on the island on 1 January 2016;

(1)

(ii) predict the population of rabbits on 1 January 2021.

(1)

(b) Use David's model to find the value of t when $R = 150$, giving your answer to three significant figures.

(2)

(c) Give **one** reason why David's model may **not** be appropriate.

(1)

(d) On the same island, the population of crickets, C , can be modelled by the formula

$$C = 1000e^{0.1t}$$

where t is the time in years after 1 January 2016.

Using the two models, find the year during which the population of rabbits first exceeds the population of crickets.

(3)
(Total 8 marks)

Q6.

Given that

$$\log_a N - \log_a x = \frac{3}{2}$$

express x in terms of a and N , giving your answer in a form not involving logarithms.

(Total 3 marks)

Q7.

(a) Sketch the graph of $y = 9^x$, indicating the value of the intercept on the y -axis.

(2)

(b) Use logarithms to solve the equation $9^x = 15$, giving your value of x to three significant figures.

(2)

(c) The curve $y = 9^x$ is reflected in the y -axis to give the curve with equation $y = f(x)$. Write down an expression for $f(x)$.

(1)

(Total 5 marks)

Q8.

(a) Given that $\log_a b = c$, express b in terms of a and c .

(1)

(b) By forming a quadratic equation, show that there is only one value of x which satisfies the equation $2 \log_2(x + 7) - \log_2(x + 5) = 3$.

(6)

(Total 7 marks)

Q9.

(a) Sketch the graph of $y = \frac{1}{2^x}$, indicating the value of the intercept on the y -axis.

(2)

(b) Use logarithms to solve the equation $\frac{1}{2^x} = \frac{5}{4}$, giving your answer to three significant figures.

(3)

(c) Given that

$$\log_a(b^2) + 3 \log_a y = 3 + 2 \log_a \left(\frac{y}{a}\right)$$

express y in terms of a and b .

Give your answer in a form not involving logarithms.

(5)

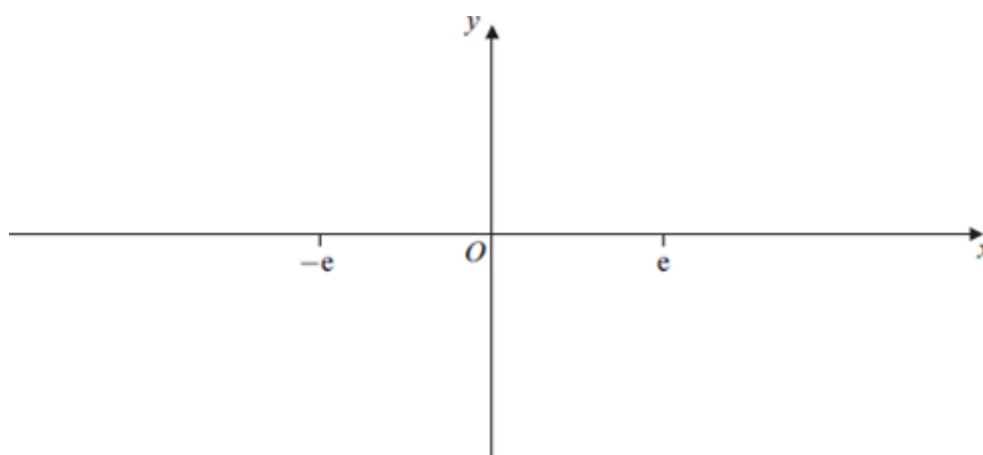
(Total 10 marks)

Q10.

- (a) Describe a sequence of two geometrical transformations that maps the graph of $y = \ln x$ onto the graph of $y = 4 \ln(x - e)$.

(4)

- (b) Sketch, on the axes given below, the graph of $y = |4 \ln(x - e)|$, indicating the exact value of the x -coordinate where the curve meets the x -axis.



(3)

- (c) (i) Solve the equation $|4 \ln(x - e)| = 4$.

(3)

- (ii) Hence, or otherwise, solve the inequality $|4 \ln(x - e)| \geq 4$.

(3)

(Total 13 marks)

Q11.

- (a) Sketch the curve with equation $y = 7^x$, indicating the coordinates of any point where the curve intersects the coordinate axes.

(2)

- (b) The curve C_1 has equation $y = 7^x$.
The curve C_2 has equation $y = 7^{2x} - 12$.

- (i) By forming and solving a quadratic equation, prove that the curves C_1 and C_2 intersect at exactly one point. State the y -coordinate of this point.

(4)

- (ii) Use logarithms to find the x -coordinate of the point of intersection of C_1 and C_2 , giving your answer to three significant figures.

(2)

(Total 8 marks)

Q12.

- (a) Write down the values of p , q and r given that:

(i) $8 = 2^p$;

(1)

(ii) $\frac{1}{8} = 2^q$;

(1)

(iii) $\sqrt{2} = 2^r$.

(1)

- (b) Find the value of x for which $\sqrt{2} \times 2^x = \frac{1}{8}$.

(2)

(Total 5 marks)

Q13.

- (a) Sketch the curve with equation $y = 4^x$, indicating the coordinates of any point where the curve intersects the coordinate axes.

(2)

- (b) Describe the geometrical transformation that maps the graph of $y = 4^x$ onto the graph of $y = 4^x - 5$.

(2)

- (c) (i) Use the substitution $Y = 2^x$ to show that the equation $4^x - 2^{x+2} - 5 = 0$ can be written as $Y^2 - 4Y - 5 = 0$.

(2)

- (ii) Hence show that the equation $4^x - 2^{x+2} - 5 = 0$ has only one real solution.

Use logarithms to find this solution, giving your answer to three decimal places.

(4)

(Total 10 marks)

Q14.

- (a) Given that $3 \ln x = 4$, find the exact value of x . (1)

- (b) By forming a quadratic equation in $\ln x$, solve $3 \ln x + \frac{20}{\ln x} = 19$, giving your answers for x in an exact form. (5)

(Total 6 marks)

Q15.

- (a) Find the value of x in each of the following:

(i) $\log_9 x = 0$; (1)

(ii) $\log_9 x = \frac{1}{2}$. (1)

- (b) Given that

$$2 \log_a n = \log_a 18 + \log_a (n - 4)$$

find the possible values of n .

(5)
(Total 7 marks)

Q16.

- (a) Sketch the curve with equation $y = 2^x$, indicating the coordinates of any point where the curve intersects the coordinate axes. (2)

- (b) (i) Use the trapezium rule with five ordinates (four strips) to find an approximate value for $\int_0^2 2^x \, dx$, giving your answer to three significant figures. (4)

- (ii) State how you could obtain a better approximation to the value of the integral using the trapezium rule. (1)

- (c) Describe a geometrical transformation that maps the graph of $y = 2^x$ onto the graph of $y = 2^{x+7} + 3$. (3)

- (d) The curve $y = 2^{x+k} + 3$ intersects the y -axis at the point $A(0, 8)$.

Show that $k = \log_m n$, where m and n are integers.

(2)

(Total 12 marks)

Q17.

A botanist is investigating the rate of growth of a certain species of toadstool. She observes that a particular toadstool of this type has a height of 57 millimetres at a time 12 hours after it begins to grow.

$$h = A \left(1 - e^{-\frac{1}{4}t} \right)$$

She proposes the model, where A is a constant, for the height h millimetres of the toadstool, t hours after it begins to grow.

(a) Use this model to:

(i) find the height of the toadstool when $t = 0$;

(1)

(ii) show that $A = 60$, correct to two significant figures.

(2)

$$h = 60 \left(1 - e^{-\frac{1}{4}t} \right)$$

(b) Use the model to:

(i) show that the time T hours for the toadstool to grow to a height of 48 millimetres is given by

$$T = a \ln b$$

where a and b are integers;

(3)

(ii) show that $\frac{dh}{dt} = 15 - \frac{h}{4}$;

(3)

(iii) find the height of the toadstool when it is growing at a rate of 13 millimetres per hour.

(1)

(Total 10 marks)

Q18.

(a) Find $\frac{dy}{dx}$ when:

(i) $y = \ln(5x - 2)$;

(2)

(ii) $y = \sin 2x$.

(2)

- (b) The functions f and g are defined with their respective domains by

$$f(x) = \ln(5x - 2), \quad \text{for real values of } x \text{ such that } x \geq \frac{1}{2}$$

$$g(x) = \sin 2x, \quad \text{for real values of } x \text{ in the interval } -\frac{\pi}{4} \leq x \leq \frac{\pi}{4}$$

- (i) Find the range of f .

(2)

- (ii) Find an expression for $gf(x)$.

(1)

- (iii) Solve the equation $gf(x) = 0$.

(3)

- (iv) The inverse of g is g^{-1} . Find $g^{-1}(x)$.

(2)

(Total 12 marks)

Q19.

- (a) Write each of the following in the form $\log_a k$, where k is an integer:

(i) $\log_a 4 + \log_a 10$;

(1)

(ii) $\log_a 16 - \log_a 2$;

(1)

(iii) $3 \log_a 5$.

(1)

- (b) Use logarithms to solve the equation $(1.5)^{3x} = 7.5$, giving your value of x to three decimal places.

(3)

- (c) Given that $\log_2 p = m$ and $\log_8 q = n$, express pq in the form 2^y , where y is an expression in m and n .

(3)

(Total 9 marks)

Q20.

- (a) Given that $3e^x = 4$, find the exact value of x . (2)
- (b) (i) By substituting $y = e^x$, show that the equation $3e^x + 20e^{-x} = 19$ can be written as $3y^2 - 19y + 20 = 0$. (1)
- (ii) Hence solve the equation $3e^x + 20e^{-x} = 19$, giving your answers as exact values. (3)
- (Total 6 marks)**

Q21.

- (a) (i) Find the value of p for which $\sqrt{125} = 5^p$. (2)
- (ii) Hence solve the equation $5^{2x} = \sqrt{125}$. (1)
- (b) Use logarithms to solve the equation $3^{2x-1} = 0.05$, giving your value of x to four decimal places. (3)
- (c) It is given that
- $$\log_a x = 2(\log_a 3 + \log_a 2) - 1$$
- Express x in terms of a , giving your answer in a form not involving logarithms. (4)
- (Total 10 marks)**

Q22.

- (a) Given that $2 \ln x = 5$, find the exact value of x . (1)
- (b) Solve the equation
- $$2 \ln x + \frac{15}{\ln x} = 11$$
- giving your answers as exact values of x . (5)
- (Total 6 marks)**

Q23.

- (a) Given that

$$\log_a x = \log_a 16 - \log_a 2$$

write down the value of x .

(1)

(b) Given that

$$\log_a y = 2 \log_a 3 + \log_a 4 + 1$$

express y in terms of A , giving your answer in a form **not** involving logarithms.

(3)

(Total 4 marks)

Q24.

(a) Sketch the graph of $y = 3^x$, stating the coordinates of the point where the graph crosses the y -axis.

(2)

(b) Describe a single geometrical transformation that maps the graph of $y = 3^x$:

(i) onto the graph of $y = 3^{2x}$;

(2)

(ii) onto the graph of $y = 3^{x+1}$.

(2)

(c) (i) Using the substitution $Y = 3^x$, show that the equation

$$9^x - 3^{x+1} + 2 = 0$$

can be written as

$$(Y - 1)(Y - 2) = 0$$

(2)

(ii) Hence show that the equation $9^x - 3^{x+1} + 2 = 0$ has a solution $x = 0$ and, by using logarithms, find the other solution, giving your answer to four decimal places.

(4)

(Total 12 marks)

Q25.

David is researching changes in the selling price of houses. One particular house was sold on 1 January 1885 for £20. Sixty years later, on 1 January 1945, it was sold for £2000. David proposes a model

$$P = Ak^t$$

for the selling price, £ P , of this house, where t is the time in years after 1 January 1885

and A and k are constants.

- (a) (i) Write down the value of A .

(1)

- (ii) Show that, to six decimal places, $k = 1.079775$.

(2)

- (iii) Use the model, with this value of k , to estimate the selling price of this house on 1 January 2008. Give your answer to the nearest £1000.

(2)

- (b) For another house, which was sold for £15 on 1 January 1885, David proposes the model

$$Q = 15 \times 1.082709^t$$

for the selling price, £ Q , of this house t years after 1 January 1885. Calculate the year in which, according to these models, these two houses would have had the same selling price.

(4)

(Total 9 marks)

Q26.

- (a) Write down the value of:

- (i) $\log_a 1$;

(1)

- (ii) $\log_a a$.

(1)

- (b) Given that

$$\log_a x = \log_a 5 + \log_a 6 - \log_a 1.5$$

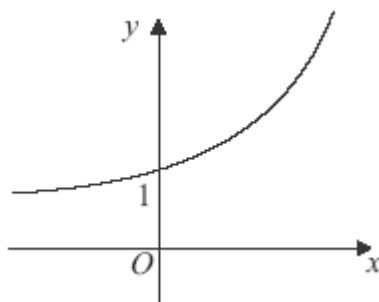
find the value of x .

(3)

(Total 5 marks)

Q27.

The diagram shows a sketch of the curve with equation $y = 6^x$.



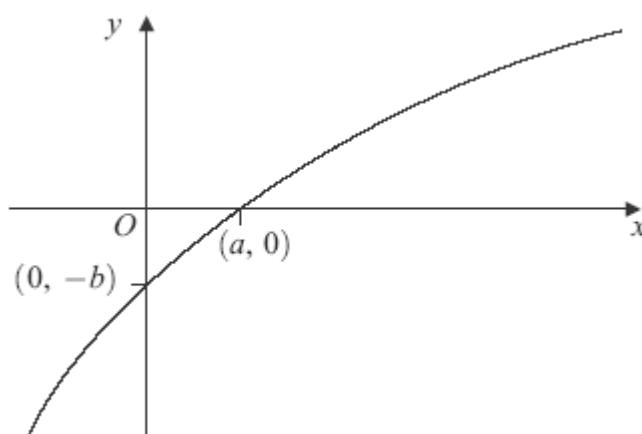
- (a) (i) Use the trapezium rule with five ordinates (four strips) to find an approximate value for $\int_0^2 6^x dx$, giving your answer to three significant figures. (4)
- (ii) Explain, with the aid of a diagram, whether your approximate value will be an overestimate or an underestimate of the true value of $\int_0^2 6^x dx$. (2)
- (b) (i) Describe a single geometrical transformation that maps the graph of $y = 6^x$ onto the graph of $y = 6^{3x}$. (2)
- (ii) The line $y = 84$ intersects the curve $y = 6^{3x}$ at the point A . By using logarithms, find the x -coordinate of A , giving your answer to three decimal places. (4)
- (c) The graph of $y = 6^x$ is translated by $\begin{bmatrix} 1 \\ -2 \end{bmatrix}$ to give the graph of the curve with equation $y = f(x)$. Write down an expression for $f(x)$. (2)

(Total 14 marks)

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Q28.

- (a) The diagram shows part of the curve with equation $y = f(x)$. The curve crosses the x -axis at the point $(, 0)$ and the y -axis at the point $(0, -b)$.



On a separate diagram, sketch the curve with the equation $y = 2f(x)$. On the diagram, indicate, in terms of a or b , the coordinates of the points where the curve crosses the coordinate axes.

(2)

- (b) (i) Describe a sequence of geometrical transformations that maps the graph of $y = \ln x$ onto the graph of $y = 4 \ln(x + 1) - 2$.

(6)

- (ii) Find the exact values of the coordinates of the points where the graph of $y = 4 \ln(x + 1) - 2$ crosses the coordinate axes.

(4)

(Total 12 marks)

Q29.

- (a) Solve the equation $3 \log_a x = \log_a 8$.

(2)

- (b) Show that

$$3 \log_a 6 - \log_a 8 = \log_a 27$$

(3)

- (c) (i) The point $P(3, p)$ lies on the curve $y = 3 \log_{10} x - \log_{10} 8$.

Show that
$$p = \log_{10} \left(\frac{27}{8} \right).$$

(2)

- (ii) The point $Q(6, q)$ also lies on the curve $y = 3 \log_{10} x - \log_{10} 8$.

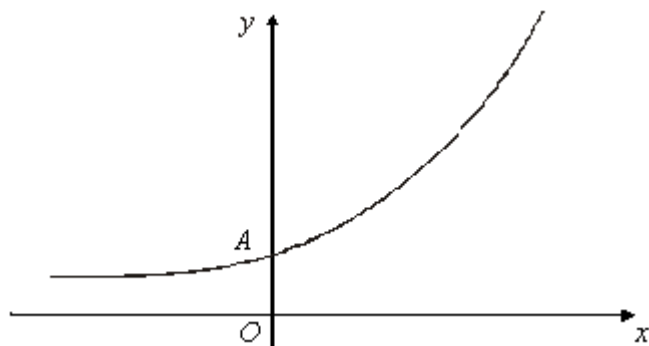
Show that the gradient of the line PQ is $\log_{10} 2$.

(4)

(Total 11 marks)

Q30.

The diagram shows a sketch of the curve with equation $y = 3(2^x + 1)$.



The curve $y = 3(2^x + 1)$ intersects the y -axis at the point A .

- (a) Find the y -coordinate of the point A . (2)
 - (b) Use the trapezium rule with four ordinates (three strips) to find an approximate value for $\int_0^6 3(2^x + 1) dx$ (4)
 - (c) The line $y = 21$ intersects the curve $y = 3(2^x + 1)$ at the point P .
 - (i) Show that the x -coordinate of P satisfies the equation $2^x = 6$ (1)
 - (ii) Use logarithms to find the x -coordinate of P , giving your answer to three significant figures. (3)
- (Total 10 marks)

Q31.

- (a) It is given that n satisfies the equation $\log_a n = \log_a 3 + \log_a(2n - 1)$
Find the value of n . (3)
- (b) Given that $\log_a x = 3$ and $\log_a y - 3 \log_a 2 = 4$:
 - (i) express x in terms of a ; (1)

- (ii) express xy in terms of a .

(4)

(Total 8 marks)

Q32.

A biologist is researching the growth of a certain species of hamster. She proposes that the length, x cm, of a hamster t days after its birth is given by

$$x = 15 - 12e^{-\frac{t}{14}}$$

- (a) Use this model to find:

- (i) the length of a hamster when it is born;

(1)

- (ii) the length of a hamster after 14 days, giving your answer to three significant figures.

(2)

- (b) (i) Show that the time for a hamster to grow to 10 cm in length is given by

$$t = 14 \ln \left(\frac{a}{b} \right), \text{ where } a \text{ and } b \text{ are integers.}$$

(3)

- (ii) Find this time to the nearest day.

(1)

- (c) (i) Show that

$$\frac{dx}{dt} = \frac{1}{14}(15 - x)$$

(3)

- (ii) Find the rate of growth of the hamster, in cm per day, when its length is 8 cm.

(1)

(Total 11 marks)

Q33.

It is given that n satisfies the equation

$$2 \log_a n - \log_a (5n - 24) = \log_a 4$$

- (a) Show that $n^2 - 20n + 96 = 0$.

(3)

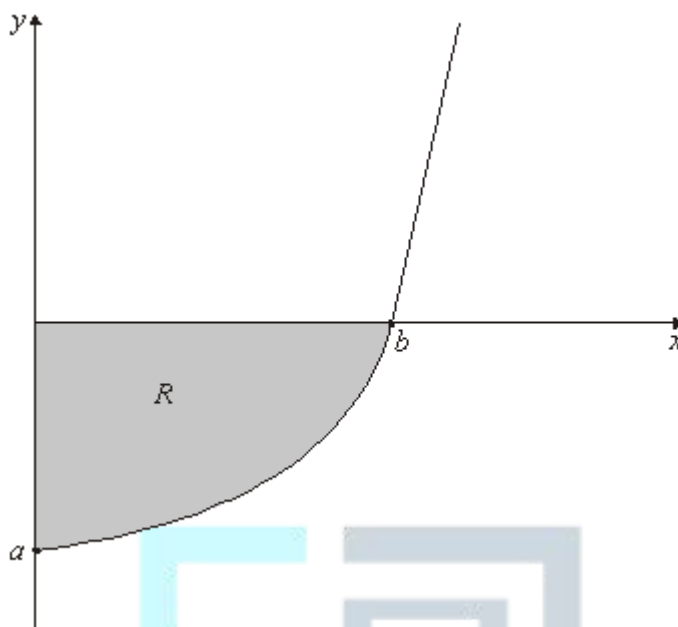
- (b) Hence find the possible values of n .

(2)

(Total 5 marks)

Q34.

The diagram shows part of the graph of $y = e^{2x} - 9$. The graph cuts the coordinate axes at $(0, a)$ and $(b, 0)$.



- (a) State the value of a , and show that $b = \ln 3$.

(3)

- (b) Show that $y^2 = e^{4x} - 18e^{2x} + 81$.

(1)

(Total 4 marks)

Q35.

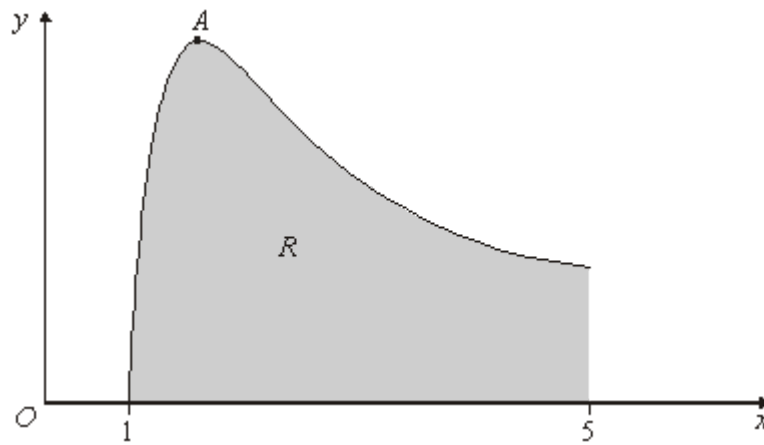
- (a) Given that $y = x^{-2} \ln x$, show that $\frac{dy}{dx} = \frac{1 - 2 \ln x}{x^3}$.

(4)

- (b) Using integration by parts, find $\int x^{-2} \ln x \, dx$.

(4)

- (c) The sketch shows the graph of $y = x^{-2} \ln x$.



- (i) Using the answer to part (a), find, in terms of e , the x -coordinate of the stationary point A .

(2)

- (ii) The region R is bounded by the curve, the x -axis and the line $x = 5$. Using your answer to part (b), show that the area of R is

$$\frac{1}{5} (4 - \ln 5)$$

(3)

(Total 13 marks)